



11892 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	UDS-1	NIRSpec MultiObject Spectroscopy	(1) uds_reduced-v4.1
	2	UDS-2	NIRSpec MultiObject Spectroscopy	(2) uds_reduced-v4.2

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	3	COSMOS-1	NIRSpec MultiObject Spectroscopy	(3) cosmos_reduced-v1.1
	4	COSMOS-2	NIRSpec MultiObject Spectroscopy	(4) cosmos_reduced-v1.2
	5	GS	NIRSpec MultiObject Spectroscopy	(5) gs_master-v1
	6	EGS	NIRSpec MultiObject Spectroscopy	(6) egs_reduced-v1

ABSTRACT

Our understanding of galaxies in the epoch of reionization has been transformed by JWST through deep imaging campaigns, wide spectroscopic surveys with low sensitivity or resolution, and high-quality spectroscopic follow-ups for small, pre-selected samples. It is now time to invest in deep medium-resolution spectroscopy for a large, unbiased sample of high-redshift sources, which will enable a statistically robust study of the general galaxy population. The JWST Single-Shutter Survey (JWS3) will obtain deep rest-frame optical spectra for a representative sample of 600 galaxies carefully selected from medium-band photometry in order to uniformly cover the redshift-magnitude space over $3 < z < 9$. The combination of deep spectroscopy and large sample size is made possible by the use of a novel observing strategy with the NIRSpec MSA, where each target is observed using a single microshutter. The JWS3 observations will have an immediate impact on our understanding of several topics: 1) the evolution of scaling relations (such as the main sequence) and their scatter, which is a direct probe of star formation burstiness; 2) the first generation of massive galaxies, via the reconstruction of their past star formation history and the detection of multi-phase outflows; 3) the incidence of supermassive black holes in the broad galaxy population and the relation between black hole mass and stellar mass. The legacy value of these observations goes well beyond the science goals listed above, making JWS3 the reference comparison sample for years to come.

OBSERVING DESCRIPTION

We observe six MSA pointings in four deep fields using the G235M and G395M gratings. Each pointing contains about 100 targets, which are galaxies at $3 < z < 9$. We assign a single microshutter to each target, to maximize multiplexing, and manually open the remaining available shutters on empty regions of the sky, which will be useful for constructing a master background spectrum. To mitigate the effect of bad pixels, we apply a 3-point dither of $(-1, 0, +1)$ shutters along the cross-dispersion direction, so that each galaxy is observed at three positions on the detector. About 20% of the targets are only observed in two out of the three dither positions due to stuck open or closed shutters.

Proposal 11892 - Targets - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	uds_reduced-v4.1	RA: 02 17 23.6676 (34.3486150d) Dec: -05 15 11.80 (-5.25328d) Equinox: J2000		
	Comments: Description=[]				
	(2)	uds_reduced-v4.2	RA: 02 17 48.3636 (34.4515150d) Dec: -05 08 57.56 (-5.14932d) Equinox: J2000		
	Comments: Description=[]				
	(3)	cosmos_reduced-v1.1	RA: 10 00 24.4219 (150.1017579d) Dec: +02 14 51.68 (2.24769d) Equinox: J2000		
	Comments: Description=[]				
	(4)	cosmos_reduced-v1.2	RA: 10 00 33.6079 (150.1400329d) Dec: +02 24 1.90 (2.40053d) Equinox: J2000		
	Comments: Description=[]				
	(5)	gs_master-v1	RA: 03 32 27.3019 (53.1137579d) Dec: -27 49 24.23 (-27.82340d) Equinox: J2000		
	Comments: Description=[]				
	(6)	egs_reduced-v1	RA: 14 19 39.2904 (214.9137100d) Dec: +52 53 3.38 (52.88427d) Equinox: J2000		
	Comments: Description=[]				

Proposal 11892 - Observation 1 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Observation	Proposal 11892, Observation 1: UDS-1											Fri Mar 13 22:04:53 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(UDS-1 (Obs 1)) Warning (Form): Config c3 (#5) has 1 primary slits affected by failed closed shutters.											
	(UDS-1 (Obs 1)) Warning (Form): Config c3 (#6) has 1 primary slits affected by failed closed shutters.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	uds_reduced-v4.1	RA: 02 17 23.6676 (34.3486150d) Dec: -05 15 11.80 (-5.25328d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	false		No	MSA Center	uds_reduced-v4.1 (3140 sources)		jwst-nirspec-mr		1.5		
Reference Stars												

Proposal 11892 - Observation 1 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G235M/F170LP)	c1		34.322321666666 67 Degrees - 5.2565336111111 38 Degrees	30.002420221530 443			1	6	8840.867
	2	2 (G395M/F290LP)	c1		34.322321666666 67 Degrees - 5.2565336111111 38 Degrees	30.002420221530 443			1	6	8840.867
	3	1 (G235M/F170LP)	c2		34.322395458333 33 Degrees - 5.2564063888888 77 Degrees	30.002413652846 148			1	6	8840.867
	4	2 (G395M/F290LP)	c2		34.322395458333 33 Degrees - 5.2564063888888 77 Degrees	30.002413652846 148			1	6	8840.867
	5	1 (G235M/F170LP)	c3		34.32246925 Degrees - 5.2562791666666 73 Degrees	30.002407084138 09			1	6	8840.867
	6	2 (G395M/F290LP)	c3		34.32246925 Degrees - 5.2562791666666 73 Degrees	30.002407084138 09			1	6	8840.867
Special Requirements	MSA Planned Aperture PA 30.0000 to 30.0000 Degrees (V3 251.42543030000002 to 251.42543030000002)										

Proposal 11892 - Observation 2 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Observation	Proposal 11892, Observation 2: UDS-2											Fri Mar 13 22:04:53 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(UDS-2 (Obs 2)) Warning (Form): Config c3 (#5) has 1 primary slits affected by failed closed shutters.											
	(UDS-2 (Obs 2)) Warning (Form): Config c3 (#6) has 1 primary slits affected by failed closed shutters.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	uds_reduced-v4.2	RA: 02 17 48.3636 (34.4515150d)									
			Dec: -05 08 57.56 (-5.14932d)									
	Equinox: J2000											
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List		Filler Candidate List		Spectral Overlap Map	Spectral Overlap Threshold	
	MSATA	false		No	MSA Center	uds_reduced-v4.2 (3541 sources)				jwst-nirspec-mr	1.5	
Reference Stars												

Proposal 11892 - Observation 2 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G235M/F170LP)	c1		34.444787625 Degrees - 5.1457569444444 29 Degrees	30.000615287298 69			1	6	8840.867
	2	2 (G395M/F290LP)	c1		34.444787625 Degrees - 5.1457569444444 29 Degrees	30.000615287298 69			1	6	8840.867
	3	1 (G235M/F170LP)	c2		34.444861374999 995 Degrees - 5.1456297222222 25 Degrees	30.000608863948 27			1	6	8840.867
	4	2 (G395M/F290LP)	c2		34.444861374999 995 Degrees - 5.1456297222222 25 Degrees	30.000608863948 27			1	6	8840.867
	5	1 (G235M/F170LP)	c3		34.444935166666 67 Degrees - 5.1455025000000 205 Degrees	30.000602436805 764			1	6	8840.867
	6	2 (G395M/F290LP)	c3		34.444935166666 67 Degrees - 5.1455025000000 205 Degrees	30.000602436805 764			1	6	8840.867
Special Requirements	MSA Planned Aperture PA 30.0000 to 30.0000 Degrees (V3 251.42543030000002 to 251.42543030000002)										

Proposal 11892 - Observation 3 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Observation	Proposal 11892, Observation 3: COSMOS-1										Fri Mar 13 22:04:53 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRSpec MultiObject Spectroscopy										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	cosmos_reduced-v1.1	RA: 10 00 24.4219 (150.1017579d) Dec: +02 14 51.68 (2.24769d) Equinox: J2000								
Acquisition	Comments: Description=[]										
	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
Template	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
Reference Stars	MSATA	false	No	MSA Center	cosmos_reduced-v1.1 (4378 sources)			jwst-nirspec-mr	1.5		

Proposal 11892 - Observation 3 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G235M/F170LP)	c1		150.10178558333 334 Degrees 2.2465041666666 665 Degrees	69.999998814087 95			1	6	8840.867
	2	2 (G395M/F290LP)	c1		150.10178558333 334 Degrees 2.2465041666666 665 Degrees	69.999998814087 95			1	6	8840.867
	3	1 (G235M/F170LP)	c2		150.10192379166 668 Degrees 2.2465544444444 445 Degrees	70.000004435751 7			1	6	8840.867
	4	2 (G395M/F290LP)	c2		150.10192379166 668 Degrees 2.2465544444444 445 Degrees	70.000004435751 7			1	6	8840.867
	5	1 (G235M/F170LP)	c3		150.10206195833 334 Degrees 2.2466047222222 22 Degrees	70.000010055684 07			1	6	8840.867
	6	2 (G395M/F290LP)	c3		150.10206195833 334 Degrees 2.2466047222222 22 Degrees	70.000010055684 07			1	6	8840.867
Special Requirements	MSA Planned Aperture PA 70.0000 to 70.0000 Degrees (V3 291.4254303 to 291.4254303)										

Proposal 11892 - Observation 4 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Observation	Proposal 11892, Observation 4: COSMOS-2											Fri Mar 13 22:04:53 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(COSMOS-2 (Obs 4)) Warning (Form): Config c1 (#1) has 1 primary slits affected by failed closed shutters. (COSMOS-2 (Obs 4)) Warning (Form): Config c1 (#2) has 1 primary slits affected by failed closed shutters. (COSMOS-2 (Obs 4)) Warning (Form): Config c3 (#5) has 1 primary slits affected by failed closed shutters. (COSMOS-2 (Obs 4)) Warning (Form): Config c3 (#6) has 1 primary slits affected by failed closed shutters. (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	cosmos_reduced-v1.2	RA: 10 00 33.6079 (150.1400329d) Dec: +02 24 1.90 (2.40053d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	false		No	MSA Center	cosmos_reduced-v1.2 (4296 sources)		jwst-nirspec-mr		1.5		
Reference Stars												

Proposal 11892 - Observation 4 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G235M/F170LP)	c1		150.141363625 Degrees 2.4017180555555 555 Degrees	70.000059047141 63			1	6	8840.867
	2	2 (G395M/F290LP)	c1		150.141363625 Degrees 2.4017180555555 555 Degrees	70.000059047141 63			1	6	8840.867
	3	1 (G235M/F170LP)	c2		150.14150183333 334 Degrees 2.4017683333333 33 Degrees	70.000065037721 85			1	6	8840.867
	4	2 (G395M/F290LP)	c2		150.14150183333 334 Degrees 2.4017683333333 33 Degrees	70.000065037721 85			1	6	8840.867
	5	1 (G235M/F170LP)	c3		150.14164004166 668 Degrees 2.4018183333333 334 Degrees	70.000071027696 07			1	6	8840.867
	6	2 (G395M/F290LP)	c3		150.14164004166 668 Degrees 2.4018183333333 334 Degrees	70.000071027696 07			1	6	8840.867
Special Requirements	MSA Planned Aperture PA 70.0000 to 70.0000 Degrees (V3 291.4254303 to 291.4254303)										

Proposal 11892 - Observation 5 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Observation	Proposal 11892, Observation 5: GS										Fri Mar 13 22:04:53 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	gs_master-v1	RA: 03 32 27.3019 (53.1137579d) Dec: -27 49 24.23 (-27.82340d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	gs_master-v1 (3972 sources)		jwst-nirspec-mr	1.5			
Reference Stars											

Proposal 11892 - Observation 5 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G235M/F170LP)	c1		53.07302075 Degrees - 27.849511388888 914 Degrees	70.018936557885 5			1	6	8840.867
	2	2 (G395M/F290LP)	c1		53.07302075 Degrees - 27.849511388888 914 Degrees	70.018936557885 5			1	6	8840.867
	3	1 (G235M/F170LP)	c2		53.073176916666 67 Degrees - 27.849461111111 11 Degrees	70.018863878537 93			1	6	8840.867
	4	2 (G395M/F290LP)	c2		53.073176916666 67 Degrees - 27.849461111111 11 Degrees	70.018863878537 93			1	6	8840.867
	5	1 (G235M/F170LP)	c3		53.073333125 Degrees - 27.849410833333 31 Degrees	70.018791179710 13			1	6	8840.867
	6	2 (G395M/F290LP)	c3		53.073333125 Degrees - 27.849410833333 31 Degrees	70.018791179710 13			1	6	8840.867
Special Requirements	MSA Planned Aperture PA 70.0000 to 70.0000 Degrees (V3 291.4254303 to 291.4254303)										

Proposal 11892 - Observation 6 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

Observation	Proposal 11892, Observation 6: EGS										Fri Mar 13 22:04:53 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(EGS (Obs 6)) Warning (Form): Config c1 (#1) has 2 primary slits affected by failed closed shutters.										
	(EGS (Obs 6)) Warning (Form): Config c1 (#2) has 2 primary slits affected by failed closed shutters.										
	(EGS (Obs 6)) Warning (Form): Config c2 (#3) has 1 primary slits affected by failed closed shutters.										
	(EGS (Obs 6)) Warning (Form): Config c2 (#4) has 1 primary slits affected by failed closed shutters.										
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	egs_reduced-v1	RA: 14 19 39.2904 (214.9137100d) Dec: +52 53 3.38 (52.88427d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	egs_reduced-v1 (2567 sources)				jwst-nirspec-mr	1.5
Reference Stars											

Proposal 11892 - Observation 6 - JWS3: Deep rest-optical spectra for an unbiased sample of galaxies at $3 < z < 9$

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G235M/F170LP)	c1		214.87830704166 666 Degrees 52.870178333333 335 Degrees	69.971725517382 86			1	6	8840.867
	2	2 (G395M/F290LP)	c1		214.87830704166 666 Degrees 52.870178333333 335 Degrees	69.971725517382 86			1	6	8840.867
	3	1 (G235M/F170LP)	c2		214.87853579166 668 Degrees 52.870228611111 11 Degrees	69.971908134929 11			1	6	8840.867
	4	2 (G395M/F290LP)	c2		214.87853579166 668 Degrees 52.870228611111 11 Degrees	69.971908134929 11			1	6	8840.867
	5	1 (G235M/F170LP)	c3		214.87876450000 002 Degrees 52.870279166666 67 Degrees	69.972090719663 36			1	6	8840.867
	6	2 (G395M/F290LP)	c3		214.87876450000 002 Degrees 52.870279166666 67 Degrees	69.972090719663 36			1	6	8840.867
Special Requirements	MSA Planned Aperture PA 70.0000 to 70.0000 Degrees (V3 291.4254303 to 291.4254303)										